

AI in Health

Hospital IT History, Modern Standards (HL7 v2, FHIR R4), and Clinical AI Modalities

Morning Theory Session: 10:00 – 12:00 | Venue: Library Studio Room

Agenda

History of Hospital IT

How healthcare evolved from paper charts and 1960s mainframes to HL7 v2 and modern FHIR R4 web standards.

- Pre-1960 Analog Paper & Tubes
- 1960s Mainframes & MUMPS
- 1980s Sockets & HL7 Delimiters
- 2014+ FHIR REST APIs

Evolution of AI Modalities

How medical data transforms into mathematical representations across four core AI paradigms.

- 1. Tabular (Classical ML)
- 2. Spatial / Images (Vision)
- 3. Time-Series (Telemetry)
- 4. Text & LLMs (RAG / MCP)

Pre-1960: How Hospitals Kept Records Before Digital

Physical charts, pneumatic vacuum tubes, and the dangers of lost medical memory

THE ANALOG HEALTHCARE SYSTEM

Paper Folders, Bed Clipboards, and Vacuum Tubes

- **The Basement Record Room:** Patient histories lived as physical paper sheets in cardboard folders filed in massive hospital basements. Retrieving a record required orderlies to manually locate and pull folders.
- **Clipboards at the Foot of the Bed:** Vital signs (temperature, pulse, blood pressure) and medication doses were hand-scribbled by nurses onto paper flowsheets hanging on patient beds.
- **Pneumatic Tube Transport:** Blood vials and paper requisition forms were loaded into brass canisters and propelled through hospital pneumatic vacuum tubes directly into the laboratory.
- **The Severe Failure Mode:** Records were frequently misfiled, stained, illegible, or lost in fires. If a patient moved to another town or hospital, their clinical history vanished completely.



The 1960s: Mainframes, Magnetic Tapes & MUMPS

The birth of digital healthcare computing and the operating system that still runs today

HARDWARE & SNEAKERNET

IBM Mainframes & Punch Cards

- **Mainframe Era:** Hospitals installed giant IBM 360/370 systems. Data was entered on 80-column Hollerith punch cards and 9-track magnetic tapes.
- **The Sneakernet:** No computer networks existed. To submit billing to Medicare or Blue Cross, technicians backed up data to tape reels, loaded them into car trunks, and physically drove them across town.
- **Centralized Processing:** Computing was strictly batch-oriented. Doctors could not view real-time records at the bedside.

SOFTWARE REVOLUTION

1966: MUMPS at Mass General

- **Created for Medicine:** Neil Pappalardo and Dr. Octo Barnett built MUMPS (Mass General Multi-Programming System) specifically for sparse medical records.
- **Hierarchical Tree Globals:** Used dynamic sparse trees (^PATIENT(id, 'LAB', 'GLUCOSE') = 105) rather than rigid relational SQL tables.
- **The Modern Legacy:** MUMPS powered the US Veterans Affairs VistA network. Today, Epic Systems (the world's largest EHR company) still runs its core backend on an advanced MUMPS engine (Caché / IRIS)!



The Medical Center's IBM 7904 Mainframe Computer System (1969)

The 1970s: Departmental Silos & Early Standards

Minicomputers fragmented the hospital into specialized islands; ASTM invented pipe delimiters

Minicomputers

DEC PDP-11 & VAX

Departments broke away from the central mainframe. Pathology, Radiology, and Pharmacy bought dedicated minicomputers running isolated software.

ASTM Committee E31

The Ancestor of HL7

ASTM standardized automated blood analyzer serial communication (ASTM E1238). Crucially, ASTM invented the pipe and caret delimiter syntax (| and ^)!

UB-82 Billing

Financial Standardization

National Uniform Billing Committee created UB-82 paper and fixed-width ASCII tape claim formats. Standardized billing codes, but zero clinical clinical depth.

Typical ASTM E1238 Message Example

```
H|\^&|||LAB_ANALYZER|||||19860512103000
P|1||P0001||DOE^JANE
O|1|ORD99|^2339-0^Glucose
R|1|^2339-0^Glucose|105|mg/dL|70-99|H||F
L|1|N
```

H (Header Segment): Initializes the transmission, establishes the dynamic delimiter set (|, \, ^, &), identifies the sending clinical instrument (LAB_ANALYZER), and stamps the exact message generation timestamp (19860512103000-> May 12, 1986 at 10:30:00 AM).

P (Patient Segment): Contains the primary administrative records for patient sequence 1, documenting the unique patient identifier (P0001) and the patient's full name separated by the component hat (DOE^JANE-> Last Name: Doe, First Name: Jane).

O (Order Segment): Tracks specific test metadata for order sequence 1, binding the lab request to an internal specimen accession/tube barcode (ORD99) and identifying the requested test code and description (^2339-0^Glucose -> LOINC 2339-0 for Glucose).

R (Result Segment): Houses the exact quantitative metric generated from the blood specimen for result sequence 1. Documents a measured value of 105 with units mg/dL evaluated against the normal clinical reference range (70-99), automatically raising an abnormal high flag (H for High) with verified final status (F for Final).

L (Message Terminator Segment): Officially marks the absolute boundary of the diagnostic transmission batch at sequence 1, confirming successful completion with termination code N (Normal termination with zero errors).

The 1980s: Local Networks & The Integration Crisis

Ethernet connected the hardware, but software was trapped in quadratic complexity

THE INTEGRATION SPAGHETTI

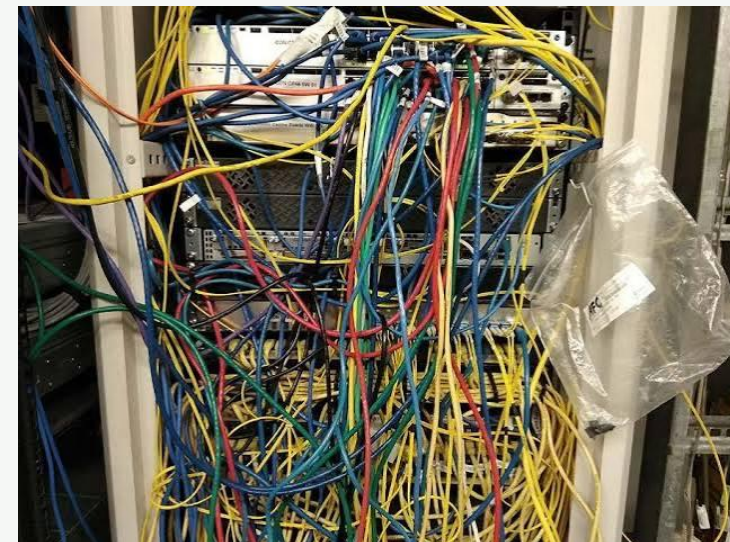
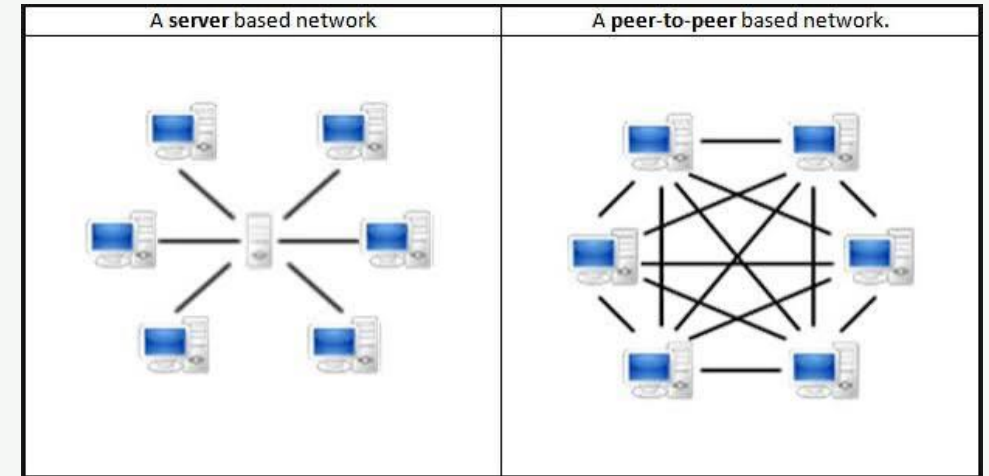
The $N(N-1)/2$ Point-to-Point Trap

- **Quadratic Explosion:** Every medical vendor used a bespoke schema. Connecting 10 departmental systems required 45 custom interfaces. Connecting 50 required 1,225 bridges!
- **ACR-NEMA 50-Pin Cable:** In 1985, CT and MRI scanners required a custom, thick 50-pin copper cable running directly from scanner to console before DICOM existed.
- **Brittle Fragility:** Upgrading a single blood analyzer broke dozens of unmaintained point-to-point translation scripts.

THE ARCHITECTURAL GAP

OSI Layers 1–4 vs. Layer 7

- **TCP/IP Solved Moving Packets:** Layers 1 through 4 (Ethernet, IP, TCP) successfully routed raw bytes from Computer A to Computer B.
- **What Do the Bytes Mean?:** When a blood machine sent packets to an EHR, the receiving software had no idea whether the payload was a patient's potassium or a billing invoice.
- **The Layer 7 Mandate:** Healthcare desperately needed an agreement at the OSI Application Layer (Layer 7). That realization sparked the creation of HL7.



1987–1989: The Birth of HL7 (Health Level Seven)

A grassroots group at the University of Pennsylvania solves hospital communication

MARCH 1987

Committee Formed at HUP

Dr. Sam Schultz (Hospital of the University of Pennsylvania) and Dr. Ed Hammond (Duke) convene hospital IT leaders to bypass slow academic standards.

OCTOBER 1987

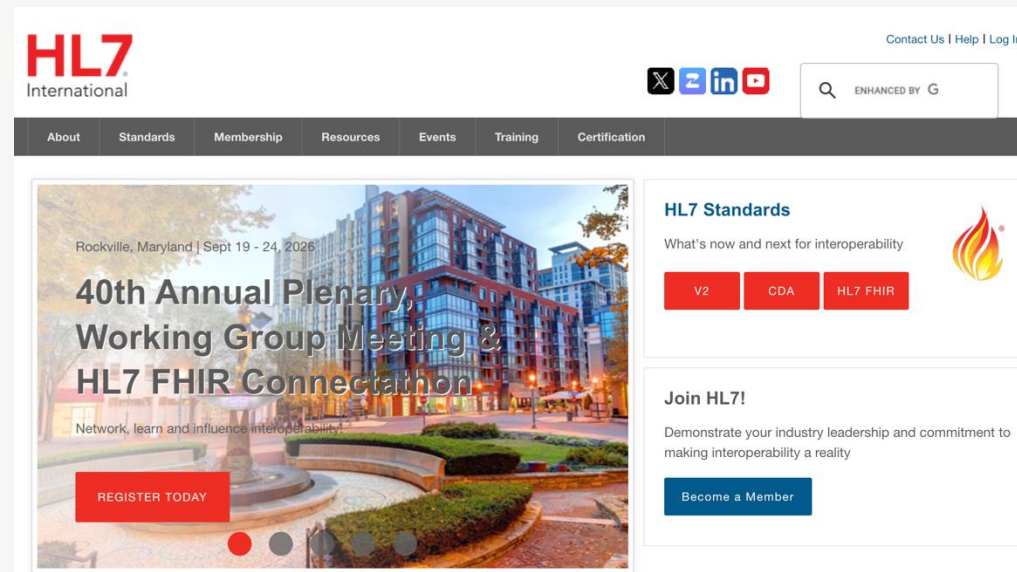
HL7 Version 1.0 (Draft)

An 84-page working draft covering basic Admission, Discharge, Transfer (ADT). Syncs patient room assignments over raw TCP sockets via MLLP framing.

1989

HL7 Version 2.1 (Commercial)

First production standard widely adopted by major commercial vendors (Baxter, SMS, Sunquest). Becomes the global workhorse of medicine.



Anatomy of an HL7 v2 Message (`ORU^R01`)

The pipe-and-hat (| and ^) ASCII format powering hospital messaging since 1989

Real HL7 v2.5 ORU^R01 (Lab Result Transmission)

```
MSH|^~\&|LAB_ANALYZER|HOSP|EMR|HOSP|19890512103000||ORU^R01|MSG01|P|2.1  
PID|1||P0001^^^MRN||DOE^JANE||19800512|F  
OBR|1|ORD99||2339-0^Glucose^LN|||19890512100000  
OBX|1|NM|2339-0^Glucose^LN|1|105|mg/dL|70-99|H|||F
```

MSH (Message Header Segment): Initializes the transmission, establishes the encoding delimiters (|, ^, ~, \, &), identifies the sending laboratory device (LAB_ANALYZER) and facility (HOSP), directs the message to the receiving clinical record system (EMR), stamps the message creation time (19890512103000 -> May 12, 1989 at 10:30:00 AM), declares the trigger event (ORU^R01 for an unsolicited clinical observation report), assigns a unique transaction ID (MSG01), sets production mode (P), and specifies the standard version (2.1).

PID (Patient Identification Segment): Contains the primary administrative and demographic records for patient sequence 1, documenting the patient's medical record number (P0001^^^MRN), legal name separated by the component hat (DOE^JANE -> Last Name: Doe, First Name: Jane), date of birth (19800512 \$> May 12, 1980), and biological sex (F for Female).

OBR (Observation Request Segment): Tracks clinical order metadata for request sequence 1, binding the laboratory order to a specific specimen accession/tube number (ORD99), identifying the universal service requested via standard coding (2339-0^Glucose^LN -> LOINC code 2339-0 for Blood Glucose), and documenting the blood sample collection timestamp (19890512100000 -> May 12, 1989 at 10:00:00 AM).

OBX (Observation / Result Segment): Houses the exact quantitative metric generated from the specimen for result sequence 1. The NM flag designates a numeric format, recording the measured observation (2339-0^Glucose^LN) with an exact value of 105 and units mg/dL evaluated against the normal reference interval (70-99), automatically raising an abnormal high alert flag (H for High) with verified final diagnostic status (F for Final).

Technological Relativity: Why HL7 v2 in 1989

An engineering triumph for 1989 hardware that accumulated 30 years of Z-segment integration debt

The Context of 1989

Why Not JSON / REST?

The Web didn't exist (HTTP came in 1991, JSON in 2001). Hardware was 1–4 MB RAM and 10 Mbps coax. Delimiter-separated ASCII allowed O(1) streaming directly into C buffers over raw TCP sockets without memory bloat.

The 'Z-Segment' Trap

Non-Standard Custom Extensions

HL7 v2 was so loosely specified that vendors added proprietary custom segments (ZBE, ZPD, ZAL). A message from Hospital A could not be parsed by Hospital B without bespoke translation code.

The Modern Debt

Point-to-Point Socket Piping

HL7 v2 runs over MLLP (raw TCP sockets). There are no URLs, no HTTP status codes, no authentication standards, and no query capabilities. Required costly interface engines (Mirth, Cloverleaf).

2003: The Cautionary Tale of HL7 Version 3

The mathematically 'perfect' XML standard that software engineers completely rejected

THE DANGERS OF OVER-ENGINEERING

How Abstract Modeling Alienated Real-World Developers

- **The Reference Information Model (RIM):** HL7 attempted to create an all-encompassing, mathematically pure object-oriented model of all medicine. Everything was modeled as an Act, Entity, or Role.
- **Massive XML Overhead:** A simple patient temperature reading required thousands of lines of deeply nested XML tags, schemas, and abstract classes. Parsing was agonizingly slow on 2003 hardware.
- **The Developer Rebellion:** Software engineers refused to adopt it. It was too difficult to learn and implement without multimillion-dollar consulting contracts. Hospitals stuck with HL7 v2, cementing v3 as a famous software failure.
- **The Engineering Lesson:** In computing, pragmatic, developer-friendly designs always beat theoretically 'pure' but unusable standards.

```
<observation classCode="OBS" moodCode="EVN">
  <!-- Unique Observation Identifier (OID) -->
  <id root="2.16.840.1.113883.19.1.1" extension="OBS-19890512-001"/>

  <!-- Test Code: LOINC 2339-0 for Glucose -->
  <code code="2339-0"
        displayName="Glucose [Mass/volume] in Blood"
        codeSystem="2.16.840.1.113883.6.1"
        codeSystemName="LOINC"/>

  <statusCode code="completed"/>
  <effectiveTime value="20030512103000"/>

  <!-- Physical Quantity (PQ): Value = 105, Unit = mg/dL -->
  <value xsi:type="PQ" value="105" unit="mg/dL"/>

  <!-- Abnormal Interpretation: H = High -->
  <interpretationCode code="H"
                     displayName="High"
                     codeSystem="2.16.840.1.113883.5.83"/>

  <!-- Reference Range: 70 - 99 mg/dL -->
  <referenceRange typeCode="REFV">
    <observationRange classCode="OBS" moodCode="EVN.CRT">
      <value xsi:type="IVL_PQ">
        <low value="70" unit="mg/dL"/>
        <high value="99" unit="mg/dL"/>
      </value>
    </observationRange>
  </referenceRange>
</observation>
```

2014–Today: Enter HL7 FHIR (Built for the Web)

Grahame Grieve redesigns healthcare data around HTTP REST, JSON, and web standards

Web-Native Standards

HTTP, REST & JSON

Built entirely on modern internet protocols. Developers query standard URLs (GET /Patient/P0001) and receive clean, human-readable JSON payloads.

The '80/20 Rule'

Pragmatic Scope

FHIR standardizes only the 80% of data concepts common across all healthcare globally. The remaining 20% edge cases are handled through explicitly validated Extensions.

Normative Stability

FHIR R4 (2019)

In 2019, FHIR Release 4 achieved Normative status. Core resources (Patient, Observation, Condition) are frozen with guaranteed permanent backwards compatibility.

```
{
  "resourceType": "Observation",
  "id": "obs-glucose-001",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "laboratory",
          "display": "Laboratory"
        }
      ]
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "2339-0",
        "display": "Glucose [Mass/volume] in Blood"
      }
    ],
    "text": "Blood Glucose"
  },
  "subject": {
    "reference": "Patient/P0001",
    "display": "Jane Doe"
  },
  "effectiveDateTime": "2026-09-09T10:30:00+07:00",
  "valueQuantity": {
    "value": 105,
    "unit": "mg/dL",
    "system": "http://unitsofmeasure.org",
    "code": "mg/dL"
  },
}
```

```
"interpretation": [
  {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation",
        "code": "H",
        "display": "High"
      }
    ]
  }
],
"referenceRange": [
  {
    "low": {
      "value": 70,
      "unit": "mg/dL",
      "system": "http://unitsofmeasure.org",
      "code": "mg/dL"
    },
    "high": {
      "value": 99,
      "unit": "mg/dL",
      "system": "http://unitsofmeasure.org",
      "code": "mg/dL"
    }
  }
]
```

FHIR, The Semantic Web & KRR (Knowledge Representation)

How FHIR implements W3C Linked Data, RDF Triples, and Description Logic reasoning in production

THE SEMANTIC WEB (W3C LINKED DATA)

URIs & The RDF Knowledge Graph

- **Universal Resource Identifiers (URIs):**

Nothing in FHIR is an ambiguous string. Every concept, coding system, and extension is anchored to a globally resolvable URI (e.g. <http://loinc.org|2339-0>).

- **RDF Triples in Practice:**

FHIR references form directed graph triples: Subject (Observation/obs-01) -> Predicate (subject) -> Object (Patient/P0001). Zero duplication; infinite traversal.

- **W3C RDF/Turtle Standard:**

HL7 and W3C officially standardized the FHIR RDF specification (fhir/rdf.html). Entire hospital populations can be queried as a unified knowledge graph via SPARQL.

KRR (KNOWLEDGE REPRESENTATION & REASONING)

Description Logics & Automated Deduction

- **Formal Ontologies (SNOMED CT):**

SNOMED CT is mathematically structured in Description Logic (the EL++ profile of OWL 2). Concepts possess formal relationships and subsumption hierarchies (Is-A).

- **Automated Logical Inference:**

If an AI clinical rule states "Alert if patient has Respiratory Infection", a reasoner infers that Bacterial Pneumonia is a subclass of Respiratory Infection and fires the alert—even if those exact words never appear in the chart!

- **Pragmatic KRR Success:**

FHIR succeeded where academic Semantic Web stalled: it hid complex Description Logics under clean, friendly JSON REST APIs that web and AI developers love.

Coded Vocabularies: The Semantic Glue

FHIR provides the grammatical schema; medical terminologies provide the dictionary

LOINC

Logical Observation Identifiers

The universal dictionary for lab assays and measurements.

- Code 2339-0: Blood Glucose
- Code 21908-9: Mean Cell Radius

Becomes column headers in ML feature matrices.

SNOMED CT

Systematized Nomenclature

The global clinical terminology for findings, body sites, and procedures.

- Code 439401001: Raman spectroscopy
- Code 44054006: Ductal carcinoma

Defines precise anatomical and procedural meaning.

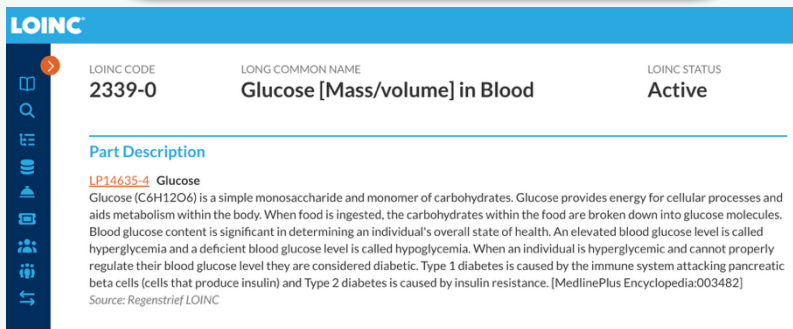
ICD-10 / ICD-11

Disease Classification

The World Health Organization standard for diagnoses and epidemiology.

- Code C50.9: Malignant breast tumor
- Code E11.9: Type 2 diabetes

Used globally for insurance claims and disease surveillance.



LOINC

LOINC CODE	LONG COMMON NAME	LOINC STATUS
2339-0	Glucose [Mass/volume] in Blood	Active

Part Description

L14635-4 Glucose

Glucose (C6H12O6) is a simple monosaccharide and monomer of carbohydrates. Glucose provides energy for cellular processes and aids metabolism within the body. When food is ingested, the carbohydrates within the food are broken down into glucose molecules. Blood glucose content is significant in determining an individual's overall state of health. An elevated blood glucose level is called hyperglycemia and a deficient blood glucose level is called hypoglycemia. When an individual is hyperglycemic and cannot properly regulate their blood glucose level they are considered diabetic. Type 1 diabetes is caused by the immune system attacking pancreatic beta cells (cells that produce insulin) and Type 2 diabetes is caused by insulin resistance. [MedlinePlus Encyclopedia:003482]

Source: Regenstrief LOINC

What is SNOMED CT?

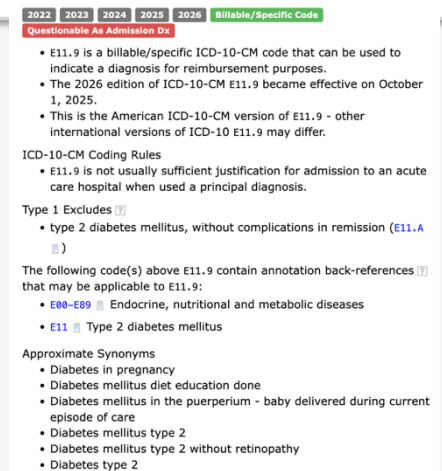
Released monthly, the SNOMED CT International Edition includes more than 360,000 concepts.

SNOMED CT:

- Is the most comprehensive, multilingual clinical healthcare terminology in the world
- Is a resource with comprehensive, scientifically validated clinical content
- Enables consistent representation of clinical content in electronic health records
- Is mapped to other international standards
- Is in use in more than eighty countries

The SNOMED CT clinical terminology has unmatched depth, enabling clinicians to record data with enhanced accuracy and consistency. SNOMED CT remains a growing and evolving product made better by the Community of Practice.

SNOMED CT supports the development of comprehensive high-quality clinical content in electronic health records. It provides a standardized way to represent clinical phrases captured by the clinician and enables automatic interpretation of these



2022 2023 2024 2025 2026 Billable/Specific Code

Questionable As Admission Dx

- E11.9 is a billable/specific ICD-10-CM code that can be used to indicate a diagnosis for reimbursement purposes.
- The 2026 edition of ICD-10-CM E11.9 became effective on October 1, 2025.
- This is the American ICD-10-CM version of E11.9 - other international versions of ICD-10 E11.9 may differ.

ICD-10-CM Coding Rules

- E11.9 is not usually sufficient justification for admission to an acute care hospital when used as a principal diagnosis.

Type 1 Excludes

- type 2 diabetes mellitus, without complications in remission (E11.A)

The following code(s) above E11.9 contain annotation back-references that may be applicable to E11.9:

- E00-E89 Endocrine, nutritional and metabolic diseases
- E11 Type 2 diabetes mellitus

Approximate Synonyms

- Diabetes in pregnancy
- Diabetes mellitus diet education done
- Diabetes mellitus in the puerperium - baby delivered during current episode of care
- Diabetes mellitus type 2
- Diabetes mellitus type 2 without retinopathy
- Diabetes type 2

Global Adoption: From Federal Law to Thailand

Where FHIR is legally mandated, and the reality of legacy hospital engines

MANDATED BY LAW

United States & Europe

- **US 21st Century Cures Act:**

Made clinical 'data blocking' illegal. Certified hospital software MUST expose standardized FHIR APIs (SMART on FHIR). Apple Health connects directly to thousands of US clinics via FHIR.

- **European Health Data Space (EHDS):**

Mandates cross-border clinical record exchange across all EU member states using FHIR.

LOCAL & REGIONAL REALITY

Thailand & Legacy Systems

- **Thailand MOPH Initiatives:**

The Ministry of Public Health and Thai health-tech agencies are actively implementing FHIR-based Health Information Exchanges (HIE) for national universal coverage digital claims.

- **The Hybrid Reality:**

Replacing hospital databases costs millions. Many provincial hospitals still run HL7 v2 and MySQL engines; FHIR is deployed as an API Gateway wrapper sitting on top.

The Four Modalities of Health AI

From 2D Tabular Feature Matrices to Spatial Tensors, Signal Telemetry, and LLM Agents

Different Math, Different Algorithms:

There is no single 'Health AI algorithm.' Tabular data requires gradient trees; radiology requires 3D convolution; telemetry requires temporal state spaces; notes require LLMs.

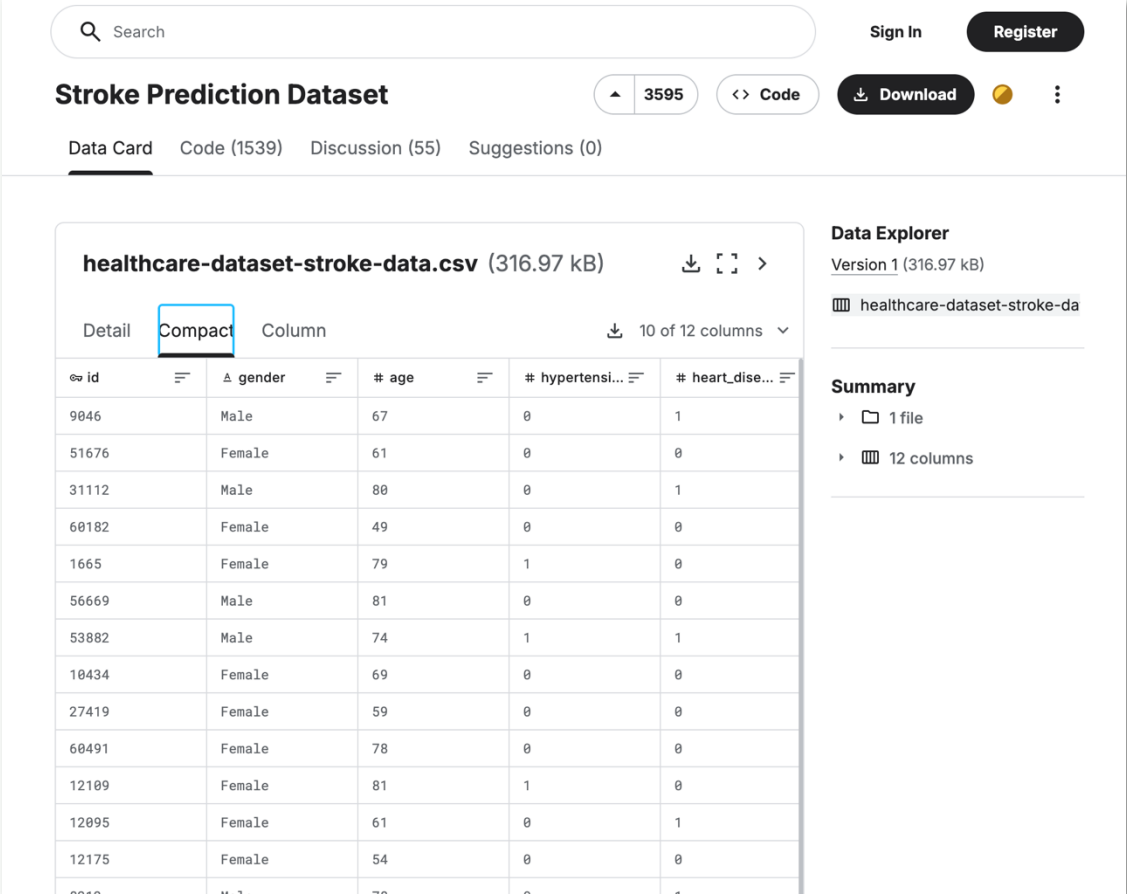
Tabular Data: The Classical Machine Learning Engine

Structured numeric matrices powering clinical risk stratification and lab assays

DATA REPRESENTATION

2D Matrices: X in $R^{N \times D}$

- **Clinical Sources:** Automated lab panels (glucose, creatinine), vital signs, cell morphometry from biopsies, and patient demographics.
- **Mathematical Structure:** A rectangular table of N patients (rows) by D features (columns). Targets y in $\{0, 1\}$.
- **Strict Requirements:** Every cell must be a valid float. Algorithms cannot handle raw nested JSON or string codes without preprocessing.



healthcare-dataset-stroke-data.csv (316.97 kB)

Detail **Compact** Column

id	gender	age	hypertension	heart_disease	stroke
9046	Male	67	0	1	
51676	Female	61	0	0	
31112	Male	80	0	1	
60182	Female	49	0	0	
1665	Female	79	1	0	
56669	Male	81	0	0	
53882	Male	74	1	1	
10434	Female	69	0	0	
27419	Female	59	0	0	
60491	Female	78	0	0	
12109	Female	81	1	0	
12095	Female	61	0	1	
12175	Female	54	0	0	

10 of 12 columns

Data Explorer
Version 1 (316.97 kB)
healthcare-dataset-stroke-da

Summary
1 file
12 columns

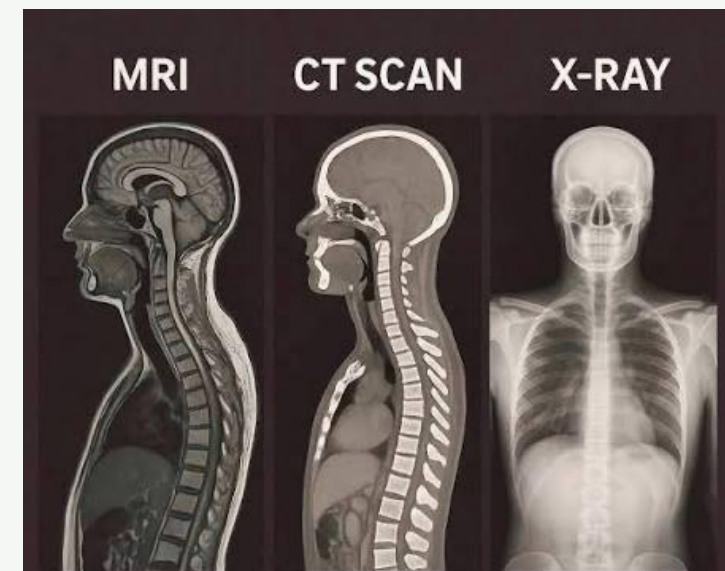
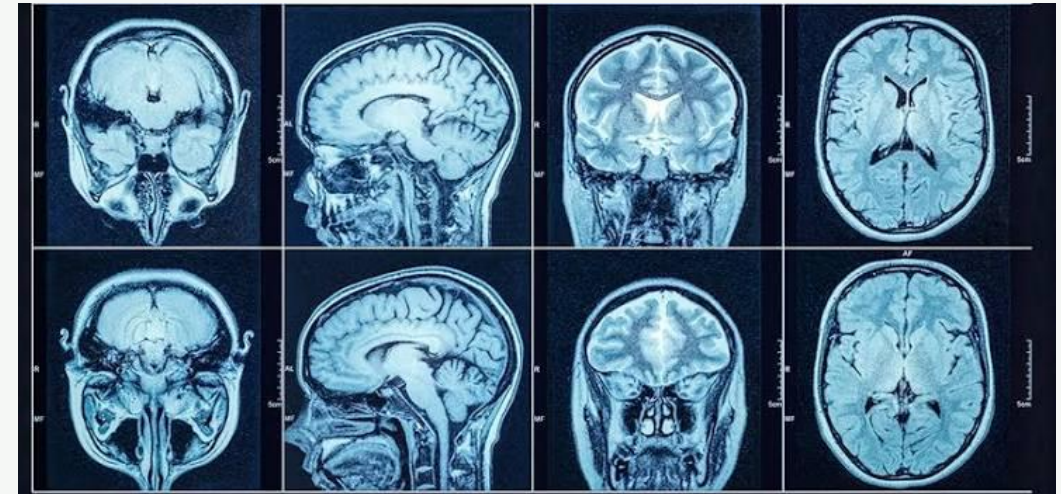
Spatial & Imaging Data: Computer Vision in Medicine

Radiographs, 3D CT/MRI scans, and gigapixel histology whole-slide images

HIGH-DIMENSIONAL SPATIAL TENSORS

From 2D X-Rays to 3D/4D Volumetric Tensors

- **The DICOM Standard:** Medical images are not simple JPEGs. They are DICOM files storing calibrated pixel arrays alongside spatial metadata (slice thickness, magnetic field strength, patient positioning).
- **Mathematical Format:** Tensors in $\mathbb{R}^{C \times D \times H \times W}$ where C = Channels/Sequences, D = Depth/Slices, H, W = Image dimensions. Whole Slide Images (WSI) reach 100,000 x 100,000 pixels.
- **AI Architectures:** Convolutional Neural Networks (ResNet, EfficientNet), Vision Transformers (ViT), and U-Net encoder-decoder models for anatomical tumor segmentation.
- **Clinical Challenge:** Multi-gigabyte file sizes, spatial resolution vs compute memory trade-offs, and requirement for expert radiologist pixel-level annotations.



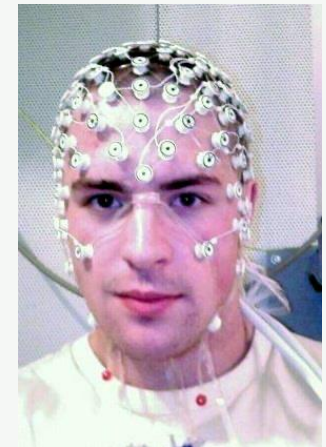
Time-Series & Telemetry: Continuous Physiological Streams

High-frequency sensor waveforms from ICU monitors, ECG leads, and wearable devices

SEQUENTIAL TEMPORAL SIGNALS

Streaming Signals at 250 Hz to 1 Sample/Hour

- **Clinical Telemetry Streams:** 12-lead Electrocardiography (ECG), continuous arterial blood pressure in ICUs, pulse oximetry (PPG), and Continuous Glucose Monitors (CGM).
- **Mathematical Representation:** 1D sequential arrays $x(t)$ sampled at high frequency. Processed via sliding windows, Fast Fourier Transforms (FFT), or Continuous Wavelet Transforms.
- **Algorithms:** Recurrent Neural Networks (LSTMs), 1D Temporal Convolutional Networks (TCN), and modern State Space Models (Mamba) for predicting acute sepsis 6 hours in advance.
- **Engineering Reality:** Sensor motion artifacts, loose electrodes, and non-stationary baseline drift require heavy digital signal processing before ML.



Unstructured Text: What 90% of People Mean by 'AI'

Physician progress notes, discharge summaries, operative logs, and clinical LLMs

THE UNSTRUCTURED MAJORITY

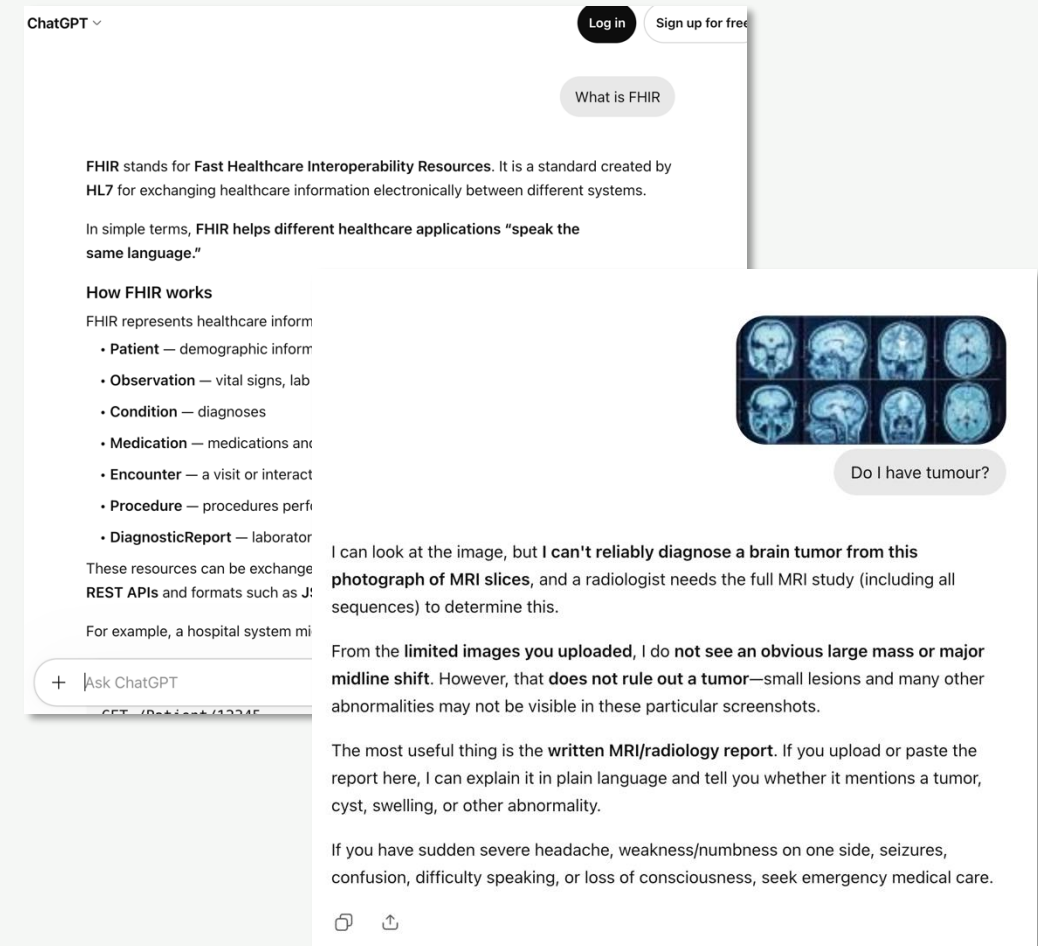
80% of Health Data Lives in Text

- **Clinical Narratives:** Doctors do not fill out structured dropdown forms; they write narrative stories explaining clinical reasoning, nuance, and history.
- **Complex Medical Jargon:** Heavy abbreviations ('sob' = shortness of breath, not an insult!), medical shorthand, and typos make rule-based parsers fail.
- **Rich Hidden Signal:** Social determinants of health (living alone, smoking history) are only documented in free-text clinical notes.

THE GENAI PARADIGM

Clinical LLMs & Copilots

- **Medical Foundation Models:** Specialized models (Med-PaLM 2, BioGPT, ClinicalBERT) trained on medical literature and clinical corpora.
- **Ambient Clinical Scribing:** Microphones listen to doctor-patient conversation in real time and automatically draft the clinical note into the EHR.
- **Clinical Question Answering:** Synthesizing complex multi-year patient records into concise diagnostic briefings for busy clinicians.



The screenshot shows a ChatGPT chat window. At the top, there are 'Log in' and 'Sign up for free' buttons. Below them is a search bar with the text 'What is FHIR'. The chat history shows a user asking 'What is FHIR?' and the AI responding with a definition of FHIR (Fast Healthcare Interoperability Resources) and a list of FHIR resources: Patient, Observation, Condition, Medication, Encounter, Procedure, and DiagnosticReport. The user then asks 'Do I have tumour?' and the AI responds with a disclaimer that it cannot reliably diagnose from a limited set of MRI slices and suggests consulting a radiologist. The user then asks 'The most useful thing is the written MRI/radiology report. If you upload or paste the report here, I can explain it in plain language and tell you whether it mentions a tumor, cyst, swelling, or other abnormality.' and the AI responds with a disclaimer that it cannot reliably diagnose from a limited set of MRI slices and suggests consulting a radiologist.

ChatGPT

Log in Sign up for free

What is FHIR

FHIR stands for **Fast Healthcare Interoperability Resources**. It is a standard created by HL7 for exchanging healthcare information electronically between different systems.

In simple terms, **FHIR helps different healthcare applications "speak the same language."**

How FHIR works

FHIR represents healthcare information in a structured way. It includes resources like:

- **Patient** — demographic information
- **Observation** — vital signs, lab results
- **Condition** — diagnoses
- **Medication** — medications and treatments
- **Encounter** — a visit or interaction
- **Procedure** — procedures performed
- **DiagnosticReport** — laboratory and diagnostic test results

These resources can be exchanged using REST APIs and formats such as JSON or XML. For example, a hospital system might use FHIR to share patient information with a specialist's office.

Do I have tumour?

I can look at the image, but I can't reliably diagnose a brain tumor from this photograph of MRI slices, and a radiologist needs the full MRI study (including all sequences) to determine this.

From the limited images you uploaded, I do not see an obvious large mass or major midline shift. However, that does not rule out a tumor—small lesions and many other abnormalities may not be visible in these particular screenshots.

The most useful thing is the **written MRI/radiology report**. If you upload or paste the report here, I can explain it in plain language and tell you whether it mentions a tumor, cyst, swelling, or other abnormality.

If you have sudden severe headache, weakness/numbness on one side, seizures, confusion, difficulty speaking, or loss of consciousness, seek emergency medical care.

The Formatting Dilemma: Why LLMs Choke on Raw FHIR

Nested JSON metadata dilutes clinical signal; Markdown tables provide the cure

THE RAW JSON TRAP

Token Bloat & Hallucination

- **Severe Token Overhead:** A single FHIR Observation consumes 300–500 tokens of nested boilerplate (resourceType, meta, system, coding). A patient chart burns 50,000+ tokens!
- **Diluted Clinical Signal:** The critical clinical number ('Glucose: 118') is buried inside hundreds of curly braces, degrading LLM attention.
- **Hallucination Risk:** High token boilerplate increases the probability that an LLM misassociates a test value with the wrong timestamp.

THE MODERN SOLUTION

Clean Markdown (.md) Tables

- **60–75% Token Reduction:** Converting FHIR resources into clean Markdown tables preserves semantic clarity while stripping thousands of redundant JSON lines.
- **Superior LLM Recall:** Foundation models are pre-trained heavily on Markdown tables and bulleted text, leading to faster inference and zero syntactic hallucinations.
- **Clean Architecture:** FHIR remains the backend database protocol; Markdown is the token-efficient presentation layer for LLM prompts.

> **explain FHIR**

- ▶ Thought for 5s, 180 tokens
Defining FHIR and Its Scope

> **{"request": "explain", "topic": "FHIR"}**

- ▶ Thought for 3s, 633 tokens
Re-evaluating FHIR Explanation Request

> >

request	topic
-----	-----
explain	FHIR

- ▶ Thought for 4s, 568 tokens
Interpreting the FHIR Explanation Request

Clinical AI Agents: RAG & Model Context Protocol (MCP)

Moving from passive chatbots to autonomous clinical copilots with deterministic tools

KNOWLEDGE RETRIEVAL

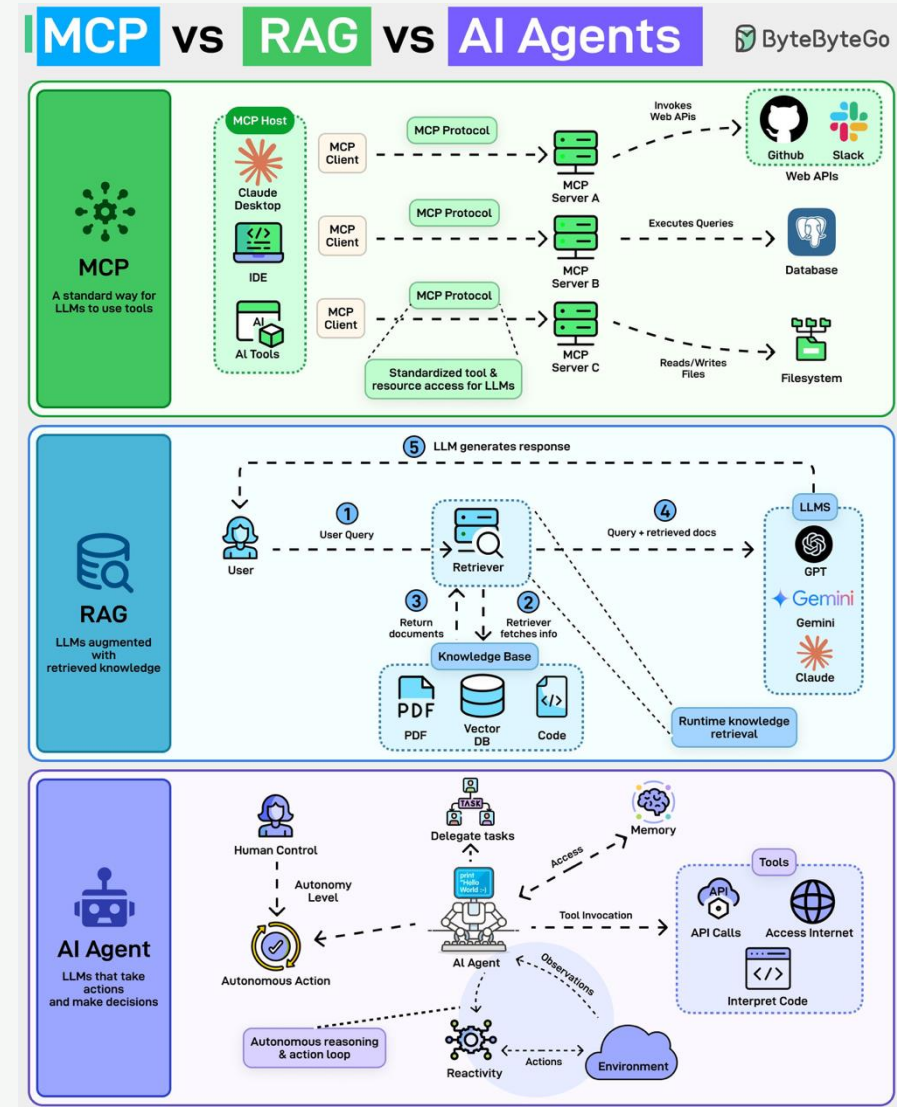
Clinical RAG & Vector DBs

- **Semantic Embeddings:** Clinical narrative notes and medical guidelines are chunked and converted into vector embeddings via clinical embedding models.
- **Vector Databases:** Stored in high-speed vector stores (pgvector, Chroma, Pinecone) for sub-millisecond similarity search.
- **Grounded Answers:** When an agent answers a question, it retrieves the top-3 most relevant patient notes and clinical trial guidelines to prevent hallucinations.

DETERMINISTIC ACTION

Model Context Protocol (MCP)

- **Tool-Calling Paradigm:** Agents do not guess; they invoke structured tools: `get_patient_labs(patient_id='P0001', code='2339-0')`.
- **Scoped & Auditable:** Every action an agent takes is logged, authenticated, and constrained by role-based access control.
- **The Future Standard:** MCP provides the open protocol for connecting AI agents to hospital databases and diagnostic instruments safely.



The Fundamental Data Impedance Mismatch

Why machine learning frameworks cannot directly ingest clinical FHIR standards

HOW HOSPITALS STORE DATA

Hierarchical Clinical Events (FHIR)

- **Data Structure:**

Deeply nested JSON resource trees linked by logical URIs and references.

- **Temporal Characteristics:**

Variable-length event histories over months or years. Irregular sampling intervals.

- **Heterogeneity:**

Mix of coded observations, narrative clinical notes, and multi-value panels.

WHAT MACHINE LEARNING REQUIRES

2D Numerical Feature Matrices

- **Data Structure:**

Rigid 2D arrays: Feature matrix X in $\mathbb{R}^{N \times D}$ and target label vector y in $\{0, 1\}$.

- **Strict Demands:**

Every patient must have exactly D columns. Algorithms like Logistic Regression cannot handle raw JSON.

- **The Workshop Core:**

Step 1 this afternoon (scripts/01_fhir_to_table.py) is the bridge that turns hierarchical FHIR into flat tabular features!

Clinical Realities: Informative Missingness & Class Imbalance

Healthcare data is not sampled at random; absence of evidence is clinical information

DATA QUALITY REALITY

Informative Missingness

- **Tests Ordered for Sick Patients:**

In sensors, missing data is hardware loss. In medicine, a troponin test or biopsy is ordered because disease is suspected! Healthy patients lack tests because they are healthy.

- **The Imputation Trap:**

Imputing missing values with population mean assigns sick values to healthy people. Must be handled deliberately via SimpleImputer inside cross-validation folds.

EVALUATION PITFALL

Class Imbalance (63% Trap)

- **The Baseline Illusion:**

In cancer screening, 63% (or 95%) of patients are benign. A dummy model predicting 'benign' every time achieves 63% accuracy while failing 100% of cancer cases!

- **Metric Requirements:**

Accuracy is dangerous. We evaluate models using Recall (Sensitivity), ROC-AUC, and tune the classification threshold deliberately.

Towards AI-Native Health Data Standards

The emerging Dual-Stack Architecture uniting legal compliance and continuous learning

THE THREE ERAS OF HEALTH DATA

From Sockets to Autonomous AI

- **Era 1: 1989–2011 (Local Machines):**

HL7 v2: Pipe-delimited ASCII strings over TCP sockets for local hospital LANs.

- **Era 2: 2014–2025 (Web & Mobile):**

HL7 FHIR: JSON REST APIs and web standards for apps, clinicians, and portals.

- **Era 3: 2026+ (Artificial Intelligence):**

AI-Native Stack: Vector embeddings, token-optimized Markdown, and Agentic MCP tool protocols.

THE DUAL-STACK SOLUTION

Record Layer + Intelligence Layer

- **Layer A · System of Record (FHIR):**

FHIR remains the authoritative, legally audited, and human-verified patient ledger for billing, medication safety, and clinical charts.

- **Layer B · System of Intelligence (AI Store):**

High-performance vector stores and feature stores (e.g. Feast, pgvector) continuously ingest FHIR streams, serving ML models and LLM agents in milliseconds.

- **Health MCP Protocol Standard:**

Standardizing tool schemas so any clinical AI agent can query and propose treatment orders across any hospital safely.

Verify your environment now so the room is 100% ready to run code at 14:00

CLOSING THE LOOP

Predictions Back as FHIR

In Step 5c of this afternoon's workshop, clicking 'Send to record' wraps our scikit-learn cancer risk prediction back into a valid FHIR RiskAssessment and Observation resource.

TERMINAL CHECK

Navigate to Repo Root

Open your terminal application and verify your current working directory:

```
$ cd fhir-ml-workshop
```

Ensure git branch is clean.

SYNC ENVIRONMENT

Verify Libraries Before Lunch

Run the fast package synchronizer from inside the repository:

```
$ uv sync
```

Verify sklearn:

```
$ uv run python -c "import sklearn;  
print(sklearn.__version__)"
```

AIT Brain Lab

Telehealth

Monitoring and Assistive Systems for Elderly and Disabled People

Asst. Prof. Chaklam Silpasuwanchai

Grounding Theory in Real AIT Systems Engineering:

Connecting optical Raman laser sensors, computer vision telemetry, and haptic rehabilitation robotics to global clinical standards and FHIR microservices.




AIT Brain

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AIT Brain Lab

The AIT Brain Lab at [Asian Institute of Technology](#) is a research lab founded by [Dr. Chaklam](#) in the year 2019.

The lab envisions to do interesting research in the intersection of human computer interaction and artificial intelligence.

[Members](#)
[Check out our projects!](#)

HIGHLIGHT NEWS

- 
ACM Trans. Health 2026: We got a paper on developing clinically validated medVQA dataset on Spines, in collaboration with IIT Kanpur and Rama 9 hospital. [Paper Link](#)
- 
CHI 2026: We got best paper award on understanding how to help humans negotiate with AI. [Paper Link](#)
- 
ACL 2026: We got a paper accepted on understanding sycophancy using personality perspective. [Paper Link](#)

Chaklam Silpasuwanchai
Home
Course ▾
AIT Brain Lab



A guy from Hong Kong. I am an Assistant Professor at the Computer Science and Information Management Program, [Asian Institute of Technology \(AIT\)](#), Thailand.

My [AIT Brain Lab](#) works in the area of Human-Centered NLP and Human-AI Interaction, primarily on understanding and modeling humans, designing health and well-being technologies and technologies that empower human lives.

I am currently an active reviewer and author for CHI, ACL, and Neurips conference. I am also a recipient of [CHI best paper award in 2017](#), and [CHI best paper award in 2026](#).

I am a recipient of a [Gates Foundation project](#) on DPI, [Toyota Mobility Foundation project](#) on crash analysis, [Ramadibodhi project](#) on spine medical images analysis, [KAKENHI Young Scientist grant](#), a [Young Scientist grant by TRF](#), a National Science and Technology Development Agency [Toray Science grant](#), and a member of a [JST grant project](#) and [NBTC project](#).

I received my Ph.D. and postdoc under the supervision of [XiangshiRen](#), [Center for Human-Engaged Computing](#), Japan.

I am co-authoring work on "[Human-Engaged Computing](#)" with [Xiangshi Ren](#), a philosophical piece discussing the need for technologies to augment humans for better survivability. This provides the philosophical ground for my research.

I am also a co-founder of a company I called "[AI Brain Lab](#)", spinning off some of our research projects to commercialization.



Chaklam Silpasuwanchai
Assistant Professor
Faculty of Technology
Asian Institute of Technology
chaklam@ait.asia

[Download CV](#)

[AIT Brain Lab zoom](#)

Research tips

- [Research process](#)
- [How to choose a good topic](#)
- [How to find related work efficiently](#)
- [What is good is NOT](#)

Area of interest

- Natural Language Processing
- Computer Vision
- Medical AI
- Brain-Computer Interfaces

Stats

- 14 Ph.D. students
- 14 Master students
- 50+ Alumni

Associated company

- AI Brain Lab

Faculty



Chaklam Silpasuwanchai
Asst. Prof. | NLP, CV, BCI

Ph.D.



Akradet Sinsamersuk
Raman Spectroscopy



Duyen Lam
Data analytics



Pranisaa Charnparttaravanit
Mental health chatbot



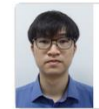
Todsavad Tangtortan
Political solipism



Arunya Prasanthu Dumunnage
Home-based stroke rehabilitation



Jirapon Kluaymaingarm
Mental health chatbot



Sarach Rujiranurak
Durian imaging



Wanchanok Sunthorn
EEG + meditation



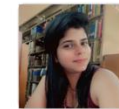
Dasuni Ganepola
Cognitive biases with AI



Mehul Parmar
Human-AI Interaction



Sila N Mahoot
Mental health chatbot



Deepali Mishra
Med VQA



Michael Lacar
Smart Rice Production



Sushmi Shrestha
AI compliance and safety

Master



Aashutosh Raut
TBD



Khin Yadanar Hlaing
Predictive Analytics



Patsakorn Tangkachaianunt
Mental health chatbot



Thet Su Sann
Speech NLP



Anubhav Kharel
Human AI



Lu Htoo Kyaw
Mentl health chatbot



Soe Htet Naing
RAG + KG for education



Vibolrottana Seng
TBD



Dechathon Niamsa-Ard
Mental Health



Min Marn Ko
Deepfake Image Detection



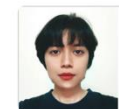
Supanut Kompayak
Face



Gaurav Karki
EEG



Nabin Gangtan Lama
Medical VQA



Theingi Win
Mental health



Home > Telehealth Monitoring and ...

Telehealth Monitoring and Assistive Systems for Elderly and Disabled People

23 Jun 2026 AIT

Share



Research Scope and Objectives

The research encompasses the development of four main work packages (WPs): non-invasive blood glucose monitoring using Raman spectroscopy (WP1), fall detection and activity monitoring via video and Wi-Fi analysis (WP2), haptic-based remote physical therapy (WP3), and a cloud-based platform for data integration and visualization (WP4). Its primary objective is to research and develop a robust telehealth infrastructure that improves safety, mobility, and independence for vulnerable populations, including the elderly, disabled, and those with chronic conditions like diabetes. Additionally, the framework lessens the burden on caregivers by enabling centralized information control, automated monitoring, and empirically supervised decision support.

WP1: Optical Glucose Sensing

Non-Invasive Raman Sensor

Measures blood glucose without skin pricks using 785nm optical Raman laser spectroscopy.

Edge ML calibration models.

WP2: Activity & Fall Telemetry

Computer Vision & Wi-Fi CSI

Monitors elderly mobility and detects falls in real time using privacy-preserving camera feeds and Wi-Fi disturbance.

Continuous spatial & time-series AI
Edge inference on embedded hardware.

WP3: Remote Haptic Therapy

Robotic Physical Rehabilitation

Upper-limb rehabilitation device providing programmable force-feedback for stroke patient recovery.

Bi-directional physical telemetry
Remote clinician tele-operation.

WP4: Cloud Telehealth Platform

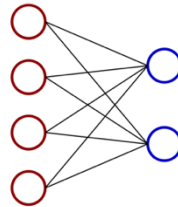
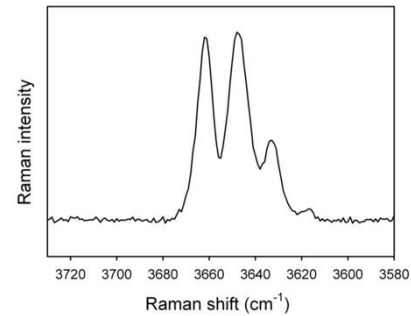
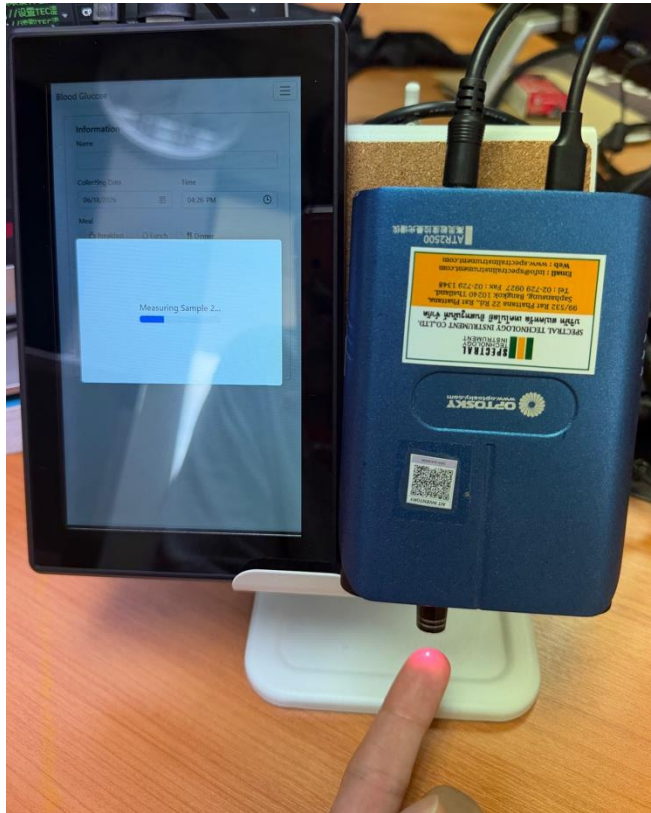
Centralize dashboard system

Consolidates telemetry from WP1–WP3 into an encrypted cloud backend for clinician review.

Real-time web & mobile dashboards.

Non-Invasive Glucose via Raman Spectroscopy and Telehealth system

WP1: Optical Glucose Sensing



118 mg/dL

POST /Observation

```
// Raman Glucose in FHIR R4 JSON
{
  "resourceType": "Observation",
  "id": "ait-raman-glucose-001",
  "status": "final",
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "2339-0",
        "display": "Glucose [Mass/vol] in Blood"
      }
    ]
  },
  "subject": { "reference": "Patient/P0001" },
  "effectiveDateTime": "2026-09-09T10:30:00+07:00",
  "valueQuantity": {
    "value": 118,
    "unit": "mg/dL",
    "system": "http://unitsofmeasure.org",
    "code": "mg/dL"
  },
  "method": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "439401001",
        "display": "In-vivo Raman spectroscopy"
      }
    ]
  }
}
```

WP4: Cloud Telehealth Platform



GET /Observation

